

MSC THESIS PRESENTATION

BEHAVIORAL FRAUD ANALYTICS

Machine Learning for Fraud Detection in Online Financial Transactions

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National & Kapodistrian University of Athens · Dept. of Economics · February 2026

3.5%

fraud rate
IEEE-CIS dataset

590K+

transactions
analyzed

What We'll Cover Today

01

Challenge & Context

The fraud detection problem and why it's hard

02

Literature & Model Choice

ML landscape review and gradient boosting rationale

03

Research Design

Four research questions guiding the study

04

Dataset & Methodology

IEEE-CIS data, pipeline, feature engineering, data split

05

Experimental Results

Four configurations across three models

06

Synthesis & Insights

Cross-experiment patterns and what they mean

07

Conclusions

Key takeaways and future research directions

Severe Class Imbalance

Fraudulent transactions represent only ~3.5% of all activity. Models can appear accurate while still failing to detect fraud reliably.

Evolving Fraud Patterns

Fraud strategies change over time, so fixed rules and manual review are limited when patterns become more complex.

False Positives vs. False Negatives

Catching more fraud usually means flagging more legitimate transactions. The operating threshold must balance fraud losses against analyst workload and customer friction.

Operational Constraints

Fraud detection systems must support timely decisions, minimize customer disruption, and remain practical for investigation workflows.

Interpretability vs. Accuracy

Ensemble models can perform strongly, but their complexity makes explainability important for understanding model decisions.

3.5%
fraud rate
in IEEE-CIS

590K+
Transactions

434
Features

3
ML Models

4
RQs

RQ 1 Model Discrimination

How effectively do gradient boosting models (XGBoost, LightGBM, CatBoost) discriminate fraudulent from legitimate transactions under ROC-AUC evaluation?

RQ 2 Impact of Downsampling

How does majority-class downsampling (1:5 ratio) affect model performance compared to training on the original imbalanced data distribution?

RQ 3 Feature Set Reduction

Can a reduced feature set, selected through cross-model feature importance agreement, preserve the discriminatory power of the full feature space?

RQ 4 Threshold Tuning Effects

How does altering the decision threshold change fraud detection behavior — specifically the recall–precision trade-off relevant to operational deployment?

A further objective is to assess the added value of engineered aggregate behavioural features such as `\_mean`, `\_rel`, `\_avg`, `\_std`, and `\_freq`, which capture historical and user-level patterns beyond raw transaction fields.

Traditional Supervised

Logistic Regression, SVM, Decision Trees, Random Forests

- Interpretable, fast, well-understood
- Requires representative labeled fraud data and special handling under severe imbalance

Anomaly Detection

K-means, DBSCAN, One-Class SVM, Isolation Forest

- Works without labeled fraud — useful when labels are scarce
- High false positive rates; not optimized for known fraud patterns in labeled data

Deep Learning

ANNs, RNN/LSTM, GNNs, Autoencoders

- Can model nonlinear, sequential, or relational fraud patterns when richer structure is available
- High-capacity, high-variance models can overfit tabular fraud data; also require large datasets, high compute, and are harder to explain

Gradient Boosting Ensembles

XGBoost, LightGBM, CatBoost

- Strong fit for structured, labeled transaction data and a practical balance of performance and feasibility
- Ensemble complexity requires SHAP or similar for interpretability

Literature takeaway: ensemble-based supervised approaches offer a favorable balance between detection performance and operational feasibility when sufficient labeled data are available.

Source: IEEE-CIS Kaggle Competition**Transaction Table**

Contains TransactionDT, TransactionAmt, ProductCD, card types, email domains, and 300+ Vesta-engineered features.

Identity Table

Contains device, browser, operating-system, and identity-related information linked to transactions.

Target Variable

Binary: isFraud = 1 for fraudulent, 0 for legitimate. Severe imbalance: ~3.5% positive rate.

Why IEEE-CIS?

Represents a large e-commerce fraud benchmark with realistic feature complexity and class imbalance. Top Kaggle solutions approached ~96% ROC-AUC.

590,540

Total Transactions

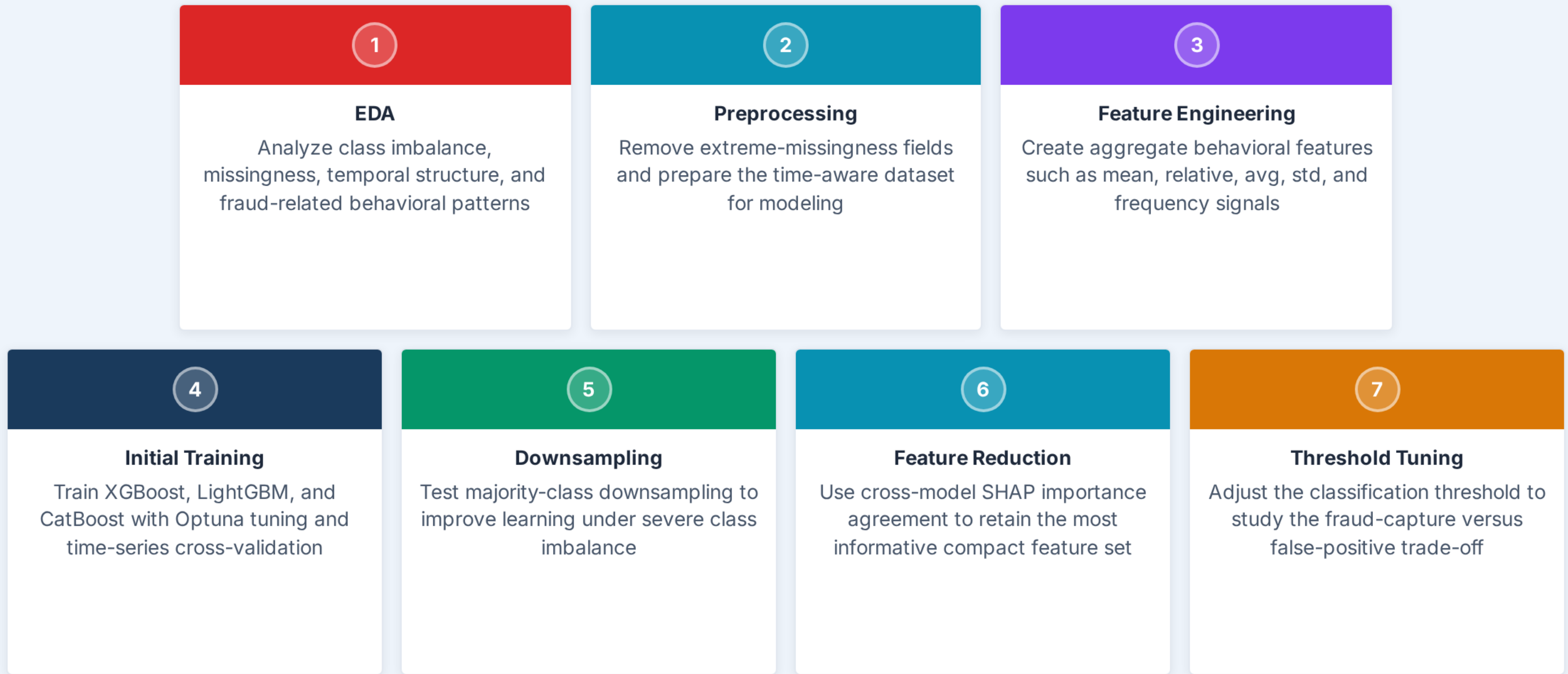
434

Feature Columns

3.5%

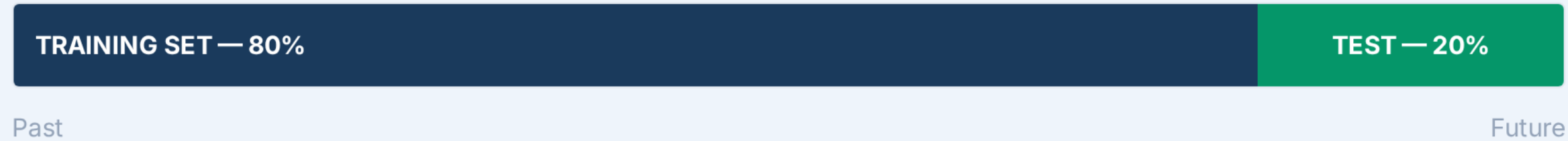
Fraud Rate (Positive)

1:5Downsample Ratio
TestedChronological data
split to simulate
real deployment



Time-series cross-validation was used throughout to prevent data leakage and simulate real-world deployment conditions.

→ Chronological order (`TransactionDT`) — no shuffling and no standalone validation split



Time-Series Cross-Validation (within training set)



Chronological evaluation is critical in fraud detection — random splitting would leak future patterns into training.

No Data Leakage

Future information never used to train on past data

80/20 Holdout

First 80% used for training, final 20% reserved as unseen test data

3-Fold CV

Validation performed inside the training set with expanding time-series folds

01 Severe Class Imbalance

Only 20,663 of 590,540 transactions are fraud (~3.5%). Accuracy alone becomes misleading, so ROC-AUC, PR-AUC, recall and threshold trade-offs matter.

02 Temporal Fraud Signatures

Raw TransactionDT looked noisy, but deriving day_of_week and hour_of_day exposed clearer fraud-rate patterns. This motivated amount-by-hour/day context features.

03 Widespread Missing Values

214 features had >50% missing values. In fraud data, missing identity/device details can itself be informative, so only >99% missing columns were dropped.

04 Behavioral & Categorical Signals

Amount extremes, 3-decimal transactions (11.77% fraud), product/card/email shifts, and rare domains justified relational aggregates: _mean, _std, _rel, _ct and _freq.

EDA TAKEAWAY: The engineered features were not arbitrary — they encoded the exact temporal, amount, user, card and category deviations surfaced by EDA.

User Anchor

- Simulated uid from card1 + addr1 + account-age proxy
- Turns anonymized rows into repeatable behavior groups
- Enables each transaction to be compared with its own baseline

Numeric Deviations

- Group numeric fields by uid, card type, product and time context
- Create _mean, _std and _rel features
- Capture transactions that are unusual for that entity/context

Categorical Frequency

- Group categorical fields by uid
- Create value-count and frequency features (_ct, _freq)
- Expose rare device, email, match-flag and identity combinations

Selection Logic

- Generate broad relational feature space first
- Let model importance remove redundant signals
- Several engineered aggregates ranked among the strongest predictors

C1

Baseline — Full Features, Imbalanced Data

Train on original data without any rebalancing. Establishes the baseline reference for raw gradient boosting.

C2

Majority-Class Downsampling (1:5 Ratio)

Reduce legitimate transactions to 1:5 ratio. Tests whether controlled rebalancing helps models learn fraud patterns better.

C3

Reduced Feature Set (SHAP-Based Selection)

Select features based on cross-model SHAP importance agreement (top 30% per model, ≥ 2 models $\rightarrow$ 215 features). Tests competitive compact models.

C4

Decision Threshold Tuning (0.1)

Lower threshold from default 0.5 to 0.1. Demonstrates the operational precision–recall trade-off.

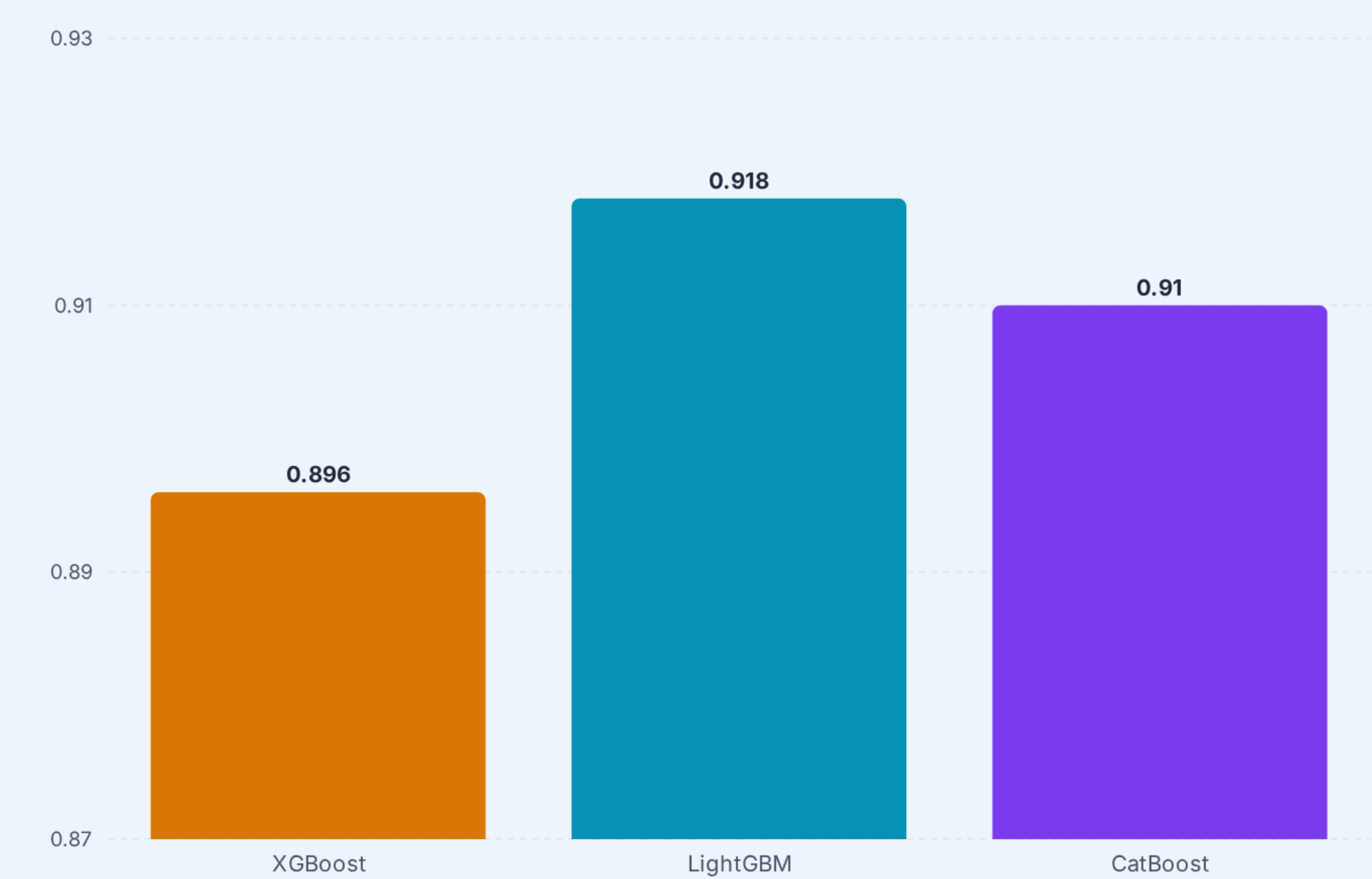
MODELS COMPARED

- XGBoost
- LightGBM
- CatBoost

EVALUATION METRICS

- ROC-AUC (primary)
- PR-AUC
- Precision / Recall
- F1-Score

LightGBM achieves the strongest baseline ROC-AUC at 0.918, ahead of CatBoost (0.910) and XGBoost (0.896)



Model	AUC	PR-AUC	F1	Prec.	Rec.
XGBoost	0.896	0.487	0.384	0.269	0.671
LightGBM	0.918	0.537	0.506	0.463	0.558
CatBoost	0.910	0.519	0.380	0.258	0.725

CatBoost shows highest recall (0.725) but low precision (0.258). LightGBM achieves the strongest baseline balance: highest AUC, PR-AUC, and F1.

Why LightGBM? Its histogram-based split finding handles the high-dimensional feature space efficiently and may act as mild regularization under severe imbalance.

Downsampling improves XGBoost and CatBoost ROC-AUC, while LightGBM keeps ROC-AUC stable and gains PR-AUC

BEFORE: Imbalanced

XGBoost

Recall: 0.671 | ROC-AUC: 0.896

LightGBM

Recall: 0.558 | ROC-AUC: 0.918

CatBoost

Recall: 0.725 | ROC-AUC: 0.910

→
1:5
DS

AFTER: 1:5 Downsampled

XGBoost

Recall: 0.703 Change +4.8% AUC: 0.908

LightGBM

Recall: 0.626 Change +12.2% AUC: 0.917

CatBoost

Recall: 0.724 Change -0.1% AUC: 0.916

Operational reading: controlled rebalancing changes recall and precision, while ROC-AUC remains strong across all three models.

Table 4: Mean |SHAP| Importance Across XGBoost, LightGBM & CatBoost

1	C1	0.422
2	C13	0.239
3	V70	0.180
4	uid_V258_mean	0.143
5	ProductCD_TransactionAmt_rel	0.141
6	uid_C1_mean	0.136
7	DeviceInfo	0.132
8	V258	0.131
9	uid_V283_mean	0.130
10	uid_C13_std	0.130
11	uid_C13_rel	0.129
12	card1_id_02_mean	0.114
13	P_emaildomain	0.107
14	uid_C13_mean	0.104
15	uid_D2_std	0.101

Global Selection

Mean absolute SHAP ranks the variables driving model predictions across the trained gradient boosting models.

Cross-Model Agreement

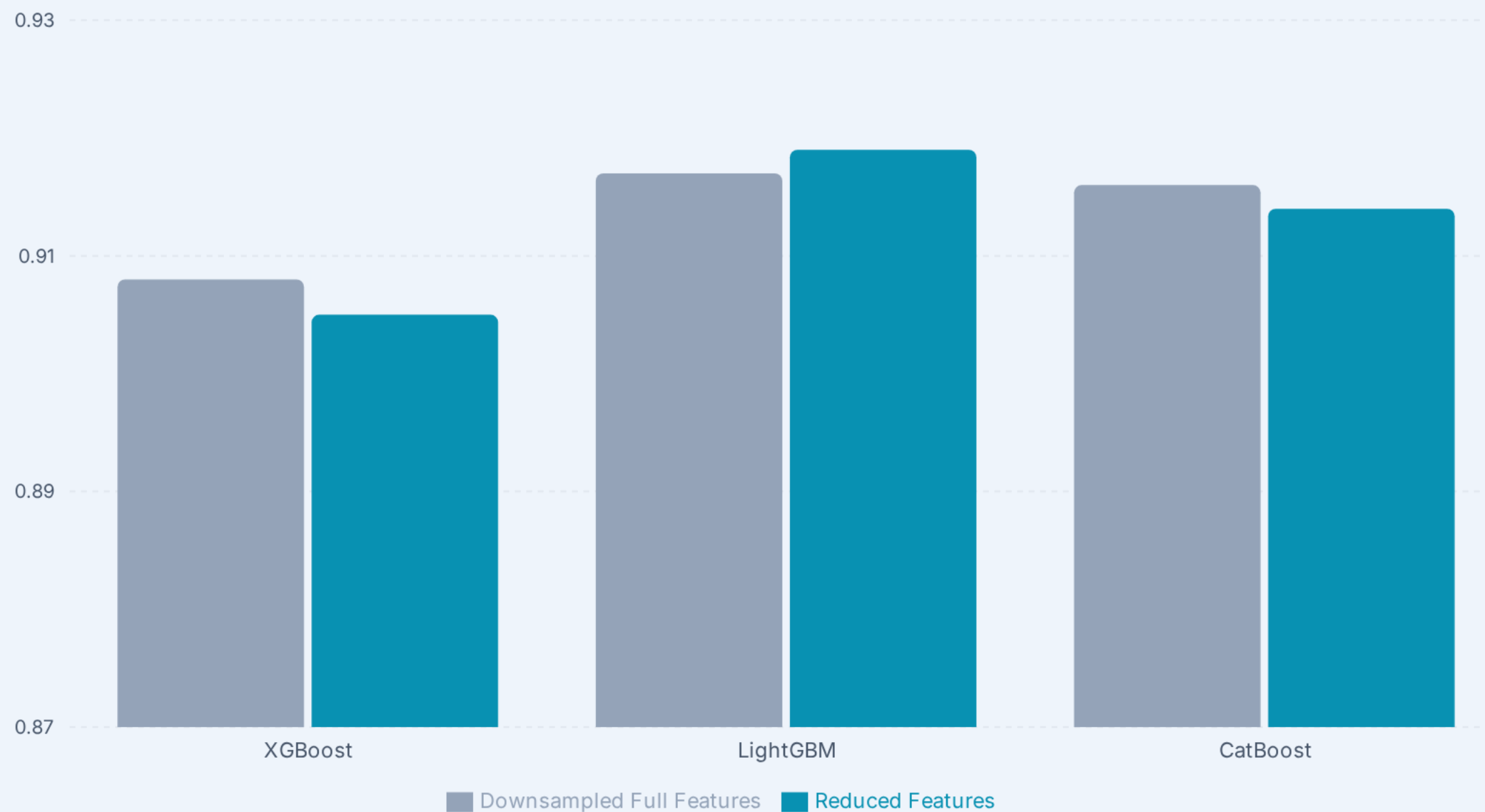
Features in the top 30% for at least two models were kept, reducing 748 model inputs to 215.

Local Explanations

The same SHAP machinery supports per-transaction attribution in the live demo.

These SHAP rankings justify the reduced-feature experiment: engineered behavioral aggregates appear alongside original C, V and identity/device fields.

Compact models remain highly competitive — LightGBM slightly improves to 0.919 ROC-AUC with reduced features



Performance Maintained

All three models retain near-identical ROC-AUC after compression. LightGBM marginally improves to 0.919.

Top Engineered Features

Behavioral aggregates (`_mean`, `_rel`, `_avg`, `_std`, `_freq`) consistently rank as the strongest predictors.

SHAP-Driven Selection

Top 30% importance per model, kept when ≥ 2 models agree — reduces model inputs from 748 to 215.

Deployment Advantage

Fewer features = faster inference, simpler pipelines, lower maintenance cost in production.

Lowering the reduced-feature model threshold to 0.1 sharply increases fraud recall at the cost of precision

XGBoost

Threshold 0.5

Recall: 0.678
Precision: 0.300



threshold → 0.1

Threshold 0.1

Recall: 0.922
Precision: 0.085

LightGBM

Threshold 0.5

Recall: 0.661
Precision: 0.348



threshold → 0.1

Threshold 0.1

Recall: 0.895
Precision: 0.110

CatBoost

Threshold 0.5

Recall: 0.720
Precision: 0.298



threshold → 0.1

Threshold 0.1

Recall: 0.938
Precision: 0.076

The thesis frames the threshold as an operational lever tied to fraud-loss tolerance and available review capacity.

CHAMPION SCORECARD

ROC-AUC Across All Configurations

Model	Baseline (Imbalanced)	Config 2 (Downsampled)	Config 3 (Reduced Features)
LightGBM	0.918	0.917	0.919
CatBoost	0.910	0.916	0.914
XGBoost	0.896	0.908	0.905

LightGBM provides the most stable ROC-AUC performance. Thesis rationale: histogram-based split finding handles the high-dimensional feature space efficiently and may regularize under severe imbalance.

What Stayed Consistent

- LightGBM was the most stable ROC-AUC performer across the main model configurations
- Engineered behavioral features repeatedly appeared as the strongest predictors
- ROC-AUC remained stable even when the experimental setup changed significantly
- Gradient boosting models proved robust to different data conditions

What Improved

- Downsampling improved XGBoost and CatBoost ROC-AUC while LightGBM gained PR-AUC
- Reduced features preserved strong discrimination with lower complexity
- Threshold reduction sharply increased fraud capture, with recall near or above 0.9
- Feature reduction preserved near-full discriminatory power with a compact feature set

What This Means

- Fraud detection is an operational optimization problem, not just a modeling task
- Model choice, sampling strategy, and threshold jointly affect deployment behavior
- ROC-AUC, PR-AUC, precision, recall and F1 each reveal different trade-offs
- A compact reduced-feature model can retain strong performance with lower complexity

Strong fraud detection depends not on the model alone, but on the full decision pipeline: features, balance, and threshold.

KEY CONCLUSIONS

- 1 Gradient boosting models are highly effective for fraud detection on the IEEE-CIS dataset, achieving strong discrimination without deep learning complexity.
- 2 LightGBM was the most dependable model overall, with ROC-AUC ≈ 0.918 – 0.919 across configurations — making it the recommended baseline choice.
- 3 Engineered aggregate behavioral features ranked among the most important predictors and helped preserve performance after feature reduction.
- 4 Threshold tuning substantially increased recall but reduced precision, showing that deployment requires explicit trade-off management.

FUTURE RESEARCH

Cost-Aware Thresholds

Select thresholds using review capacity or explicit false-positive and false-negative costs

Probability Calibration

Improve whether predicted risk scores reflect true fraud likelihood

Richer Features & Data

Use behavioral histories, temporal patterns, relational signals, and larger transaction datasets

Robustness Extensions

Explore ensemble strategies and drift-aware evaluation as fraud patterns evolve

THANK YOU

Thank you for your attention

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